



**The  
University of  
Faisalabad**

# **Computer Network**

## **Basics of Network**

**Transmission Media & Types**

**Guided Media: Twisted Pair Cable, Unshielded Twisted Pair,  
Coaxial Cable, Optical Fibers**

**Unguided Media: Radio waves, Microwaves, Infrared**

**Lecture #2**

**Week#1**

**Data Communications and Networking, Fourth Edition, Behrouz A. Forouzan**

**Prepared by:**

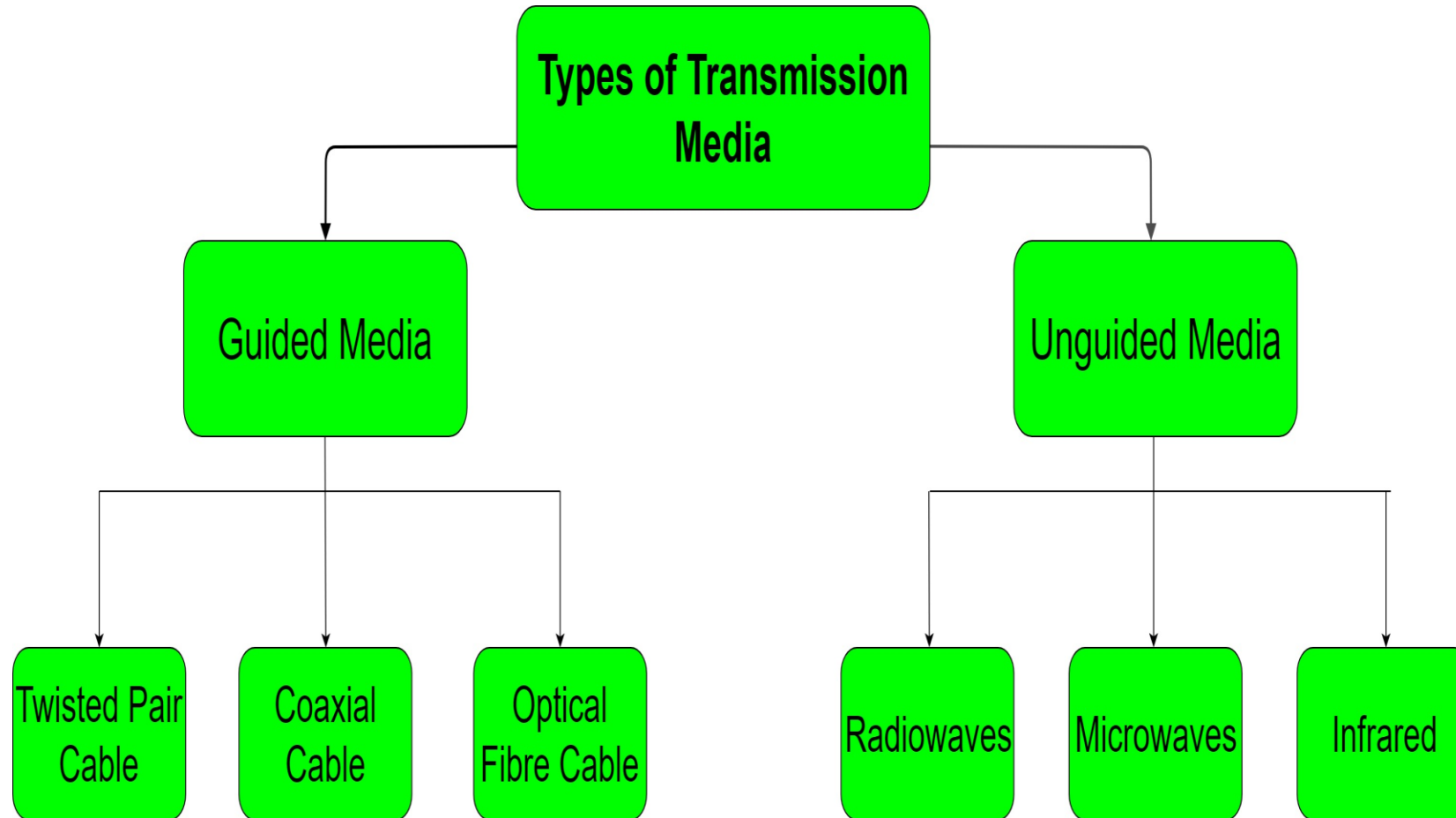
**Mehak Rana**

*Lecturer, Department of Computer Science  
The University of Faisalabad*

- In network communications, a **transmission medium** is a physical connection or an interface between the transmitter and the receiver.
- A **transmission medium** is the **path** through which data travels from one device to another.
- It acts like a **bridge** between the **transmitter** and the **receiver**.
- **What is a Transmitter?**
- The **transmitter** is the device that **sends data**
- Mobile phone sending a message
- **What is a Receiver?**
- The **receiver** is the device that **receives data**.

Mobile phone receiving a message

- There are two major categories of transmission media, namely
  - Guided Media (Wired Communication)
  - Unguided Media (Wireless Communication)



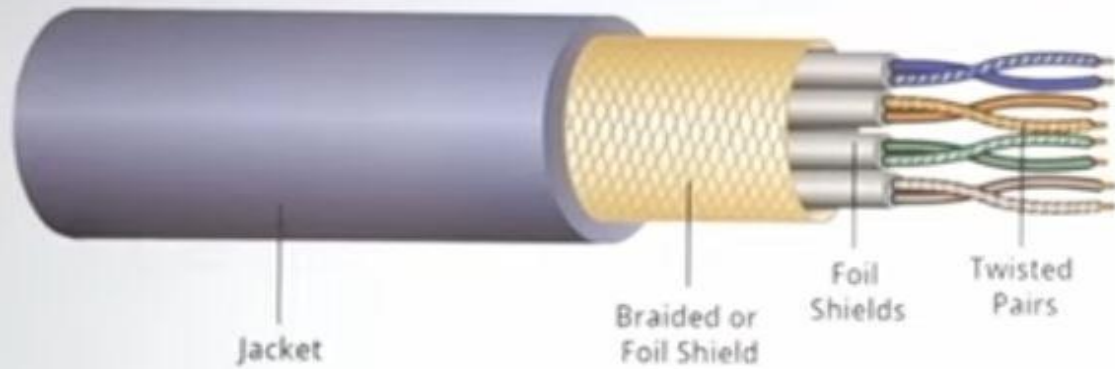
# Guided Media

- It is also referred to as Wired or Bounded transmission media.
- Signals being transmitted are directed and confined in a narrow pathway by using physical links.
- **Features:**
  - High Speed
  - Secure
  - Used for comparatively shorter distances

# Twisted Pair Cable

- It consists of 2 separately insulated conductor wires wound about each other.
- Generally, several such pairs are bundled together in a protective sheath.
- They are the most widely used Transmission Media.
- Twisted Pair is of two types:
  - **Unshielded Twisted Pair (UTP)**
  - **Shielded Twisted Pair (STP)**
- **Real-life examples:**
  - Landline telephone wires
  - Ethernet cables in homes/offices
  - Computer lab networking
  - Common in LANs (Local Area Networks).

# Twisted Pair Cable



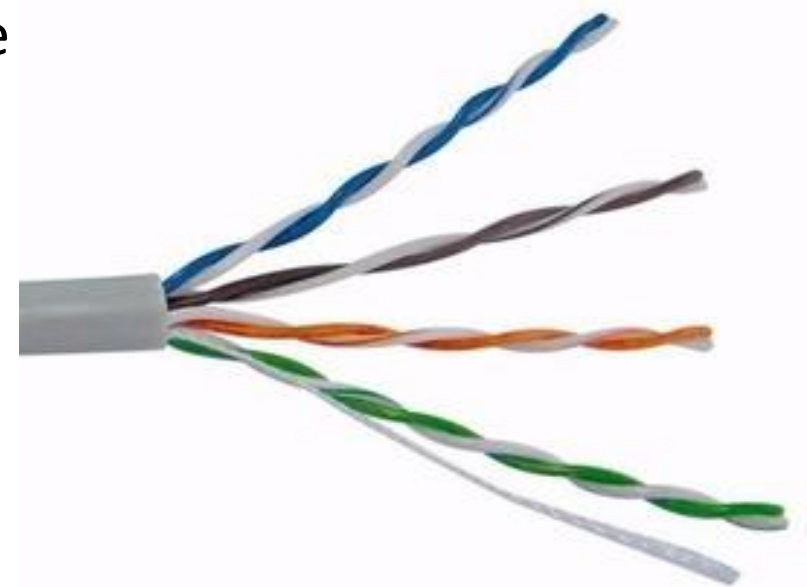
# Unshielded Twisted Pair (UTP)

- **UTP** is a type of cable made of **pairs of copper wires twisted together**. UTP is a low-cost twisted pair cable without shielding, commonly used for short-distance data transmission in computer networks.

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The twisting helps **reduce interference**, even though there

- Data travels as through the copper wires.
- It is used for telephonic applications.
- **Advantages:**
  - Least expensive
  - Easy to install
  - High speed capacity
  - Lower capacity and performance in comparison to STP
  - Short distance transmission





**STP** is a twisted pair cable in which the copper wires are **covered with an extra metallic shield**.

This shield protects the signals from **external interference and noise**.

It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

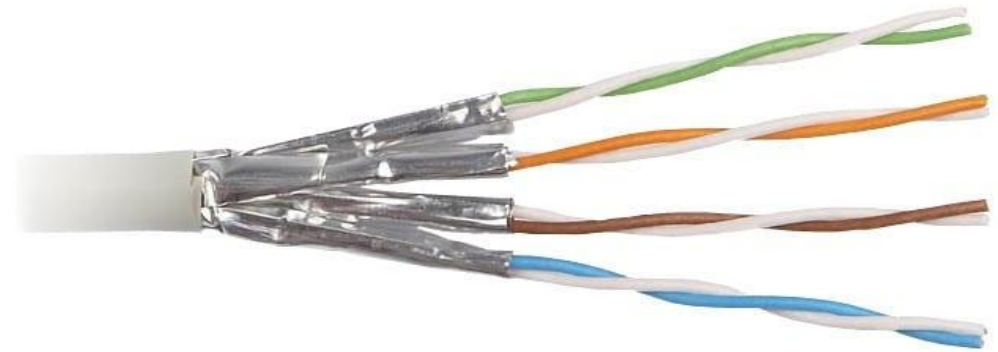
## Advantages:

- Better performance at a higher data rate in comparison to UTP
- Comparitively faster
- Comparitively difficult to install and manufacture
- More expensive

## Where is it used?

Industrial areas

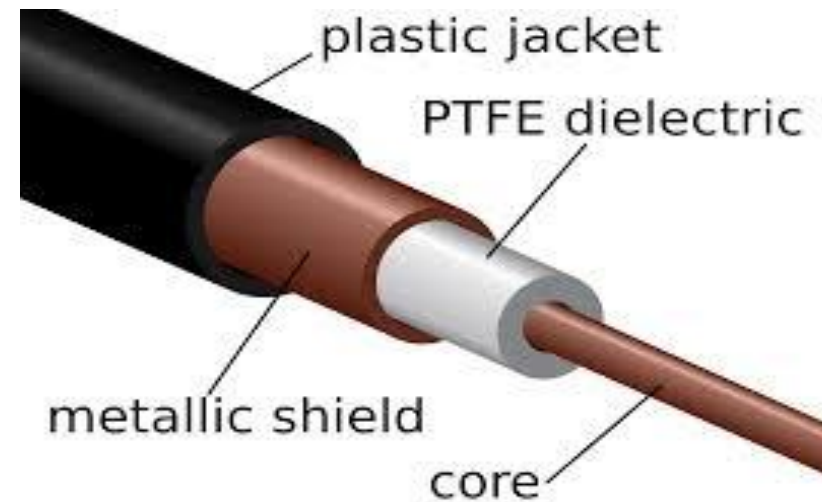
Banks networks



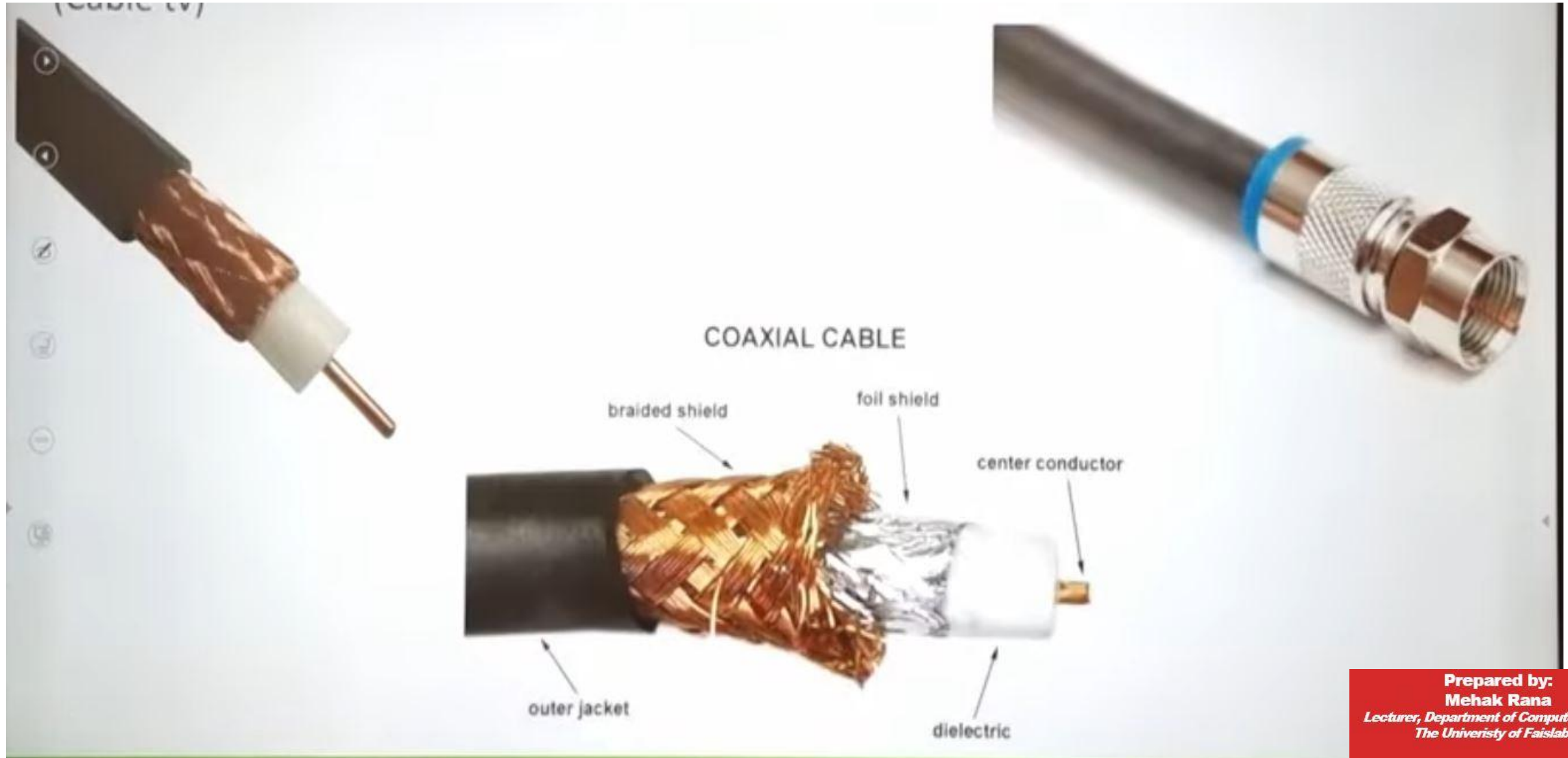


**One central copper wire** (carries data)

- **Insulation layer**
- **Metal shield** (protects from noise)
- **Outer plastic cover**
- Because of the shield, it has **better protection from noise** than twisted pair cables.
- It has an outer plastic covering containing 2 parallel conductors each having a separate insulated protection cover.
- Example:
- Cable TVs widely use Coaxial cables.
- **Advantages:**
  - High Bandwidth
  - Better noise Immunity
  - Easy to install and expand
  - Inexpensive
- **Disadvantages:**
  - Single cable failure can disrupt the entire network
- **Real-life examples:**
  - Cable TV connections
  - CCTV camera wiring
  - Better protection from noise than twisted pair.

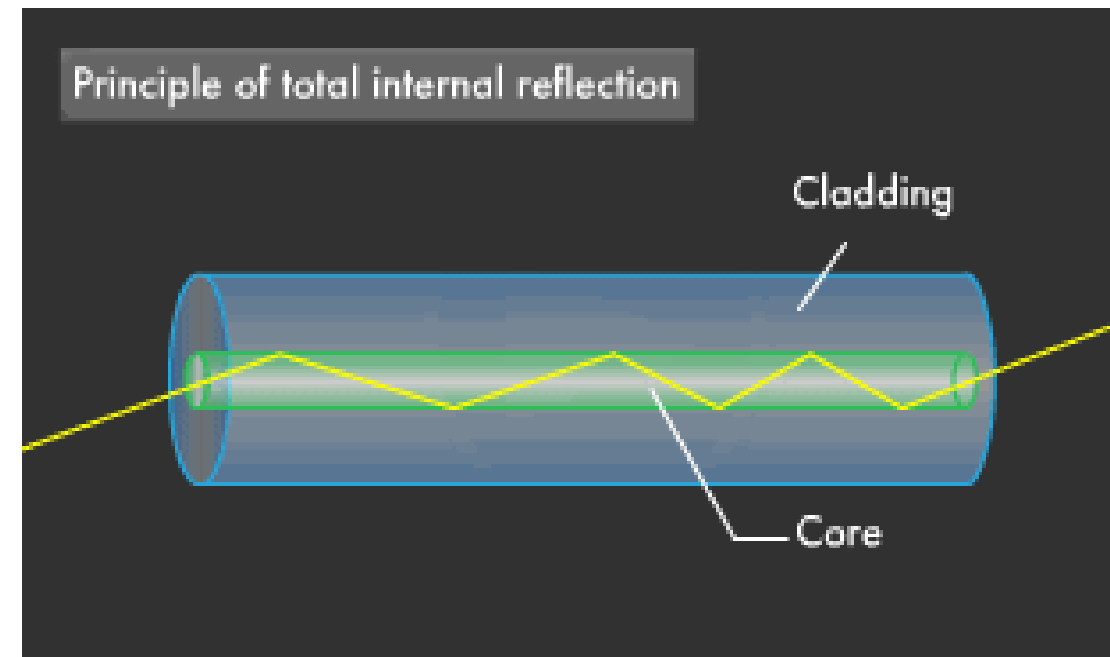


# Coaxial Cable



# Optical Fibers

- It uses the concept of reflection of light through a core made up of glass or plastic.
- The core is surrounded by a less dense glass or plastic covering called the cladding.
- It is used for transmission of large volumes of data.
- The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.
- **Structure:**
  - **Core** → carries light (glass or plastic)
  - **Cladding** → less dense layer around the core
  - **Outer jacket** → protects the fiber
  - The cladding keeps the light **inside the core** by reflection
- **Real-life examples:**
  - Fiber internet (PTCL)
  - Hospital and university high-speed networks
  - Best for long distance & high-speed data.



- It is also referred to as Wireless or Unbounded transmission media. No physical medium is required for the transmission of electromagnetic signals.
- **Features:**
  - Signal is broadcasted through air
  - Used for larger distances
- **There are 3 major types of Unguided Media**
  - Radio waves
  - Microwaves
  - Infrared



- **Radio waves** are **wireless signals** that carry data **through air**. They do **not need wires or cables**.
  - **How do radio waves work?**
  - A **transmitter** sends signals as radio waves
  - The waves travel through air
  - A **receiver** catches those waves
- Like **talking and listening**, but without wires.



## Real-life examples:

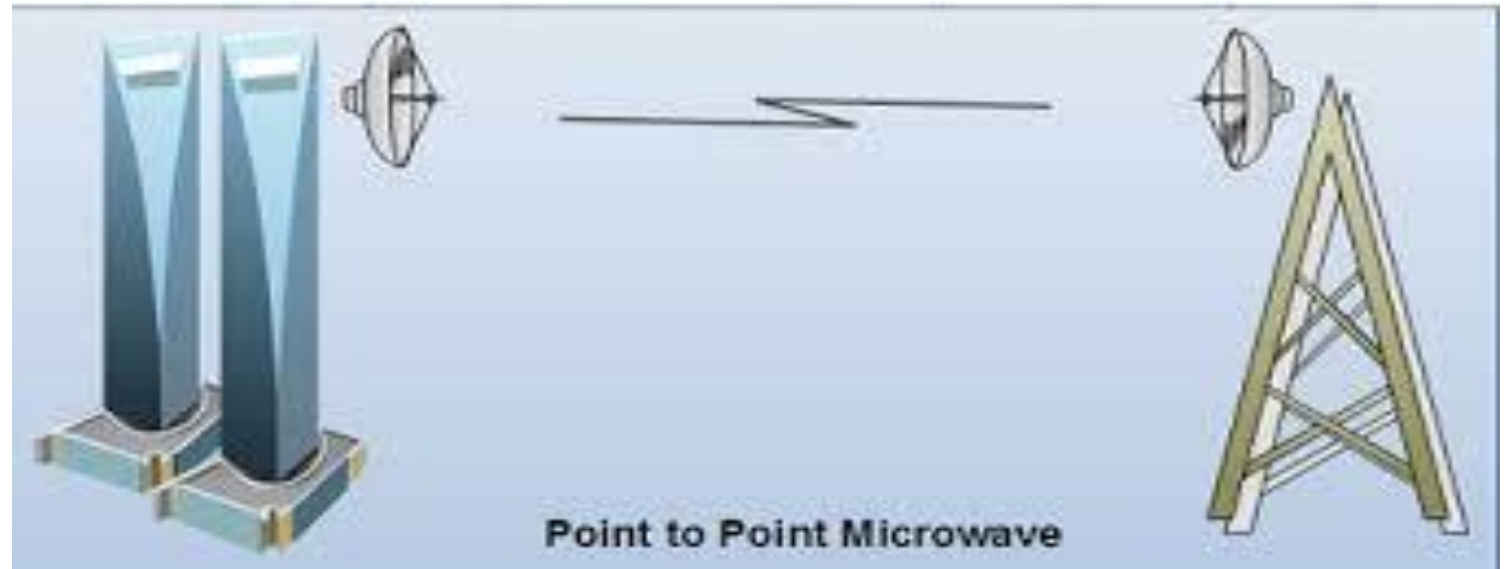
- FM radio
- Wi-Fi
- Bluetooth

- Infrared waves are used for very short distance communication.
- **How does infrared work?**
- A **transmitter** sends infrared signals
- The signal travels **in a straight line**
- A **receiver** detects the signal
- It usually needs **line of sight** (no walls in between).
- It is used in TV remotes, wireless mouse, keyboard etc.
- **Real-life examples:**
- TV remote control
- Wireless headphones
- File transfer between phones

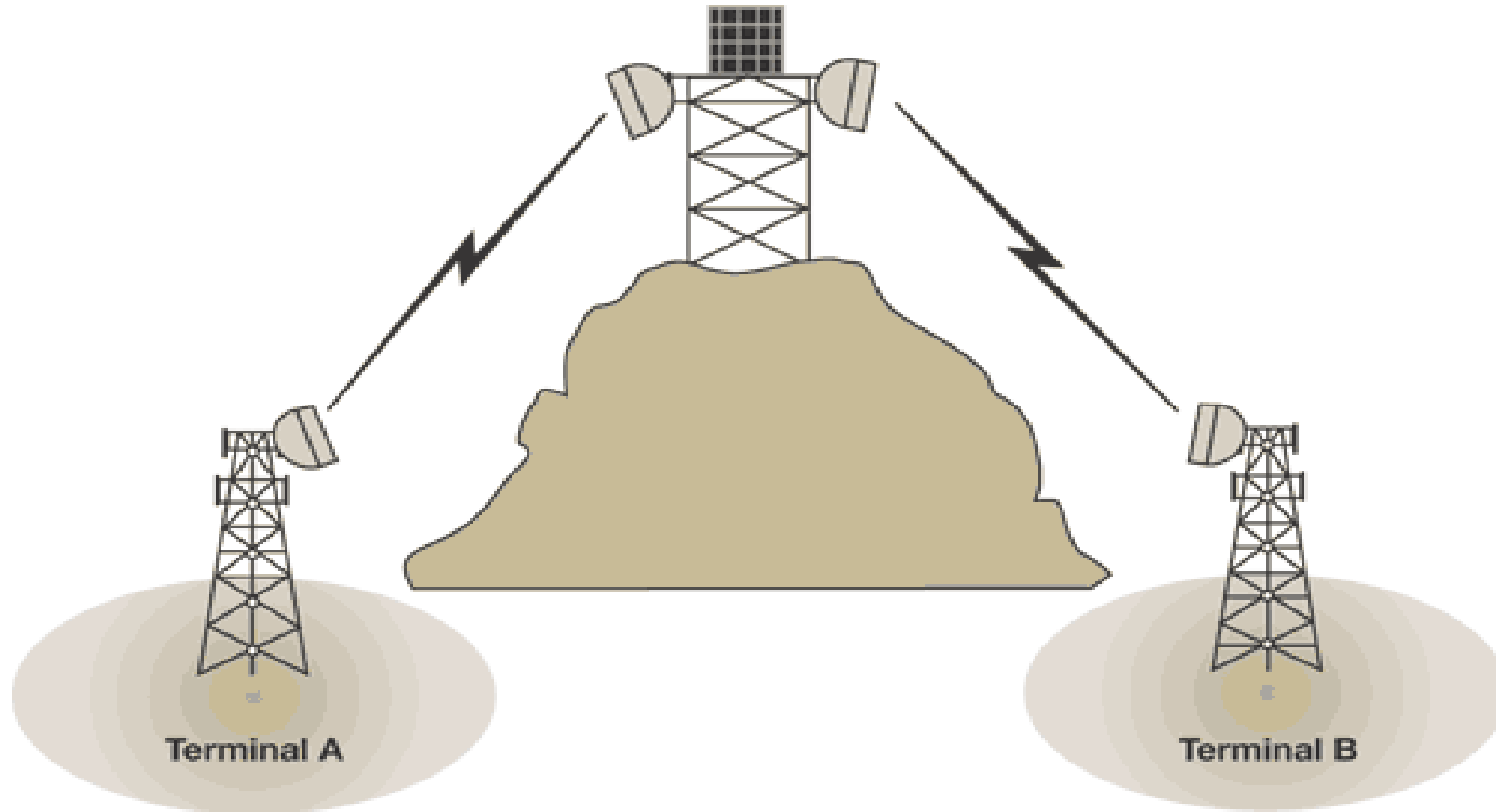




- It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other.
- The distance covered by the signal is directly proportional to the height of the antenna.
- These are majorly used for mobile phone communication and television distribution.
- **Real-life examples:**
  - Mobile Phone Towers
  - Satellite communication
  - Microwave ovens (same type of waves)
  - Used for long-distance wireless links.







**1. A microwave repeater link is designed to transfer signals from one terminal station to another without loss of traffic or signal performance.**

# THANK YOU

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## You can Find me:

mehakrana.CS@tuf.edu.pk